THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO SECONDARY SCHOOL

FORM FOUR PRE-MOCK EXAMINATION
032 CHEMISTRY

Time: 3 Hours **Friday, April 25th, 2025 a.m.**

INSTRUCTIONS

1. This paper consists of sections A, B, and C with **eleven (11)** questions.
2. Answer all questions in section A and B and **two (2)** questions from section C.
3. Section A carries **sixteen (16)** marks, Section B carries **fifty-four (54)** marks and Section C carries **thirty (30)** marks
4. Cellular phones and any unauthorized materials are not allowed in the examination room.
5. Write your **examination number** on every page of your answer sheet(s)
6. The following constants may be used:
   Atomic mass: H=1, C=12, N=14, O=16, Na=23, Ca = 40, S=32, Mg=24, Cl=35.5, Cu = 63.5, Pb = 207, Fe = 56
   Avogadro's number = $6.02 \times 10^{23,}$ GMV at S.T.P = $22.4dm^3$, 1Faraday = 96,500coulombs.
   Standard pressure = 760mm Hg, Standard temperature = 273K, 1 litre = $1dm^3 = 1000cm^3$.

SECTION A (16 MARKS)

1. For each of the items (i) – (x), choose the most correct answer from the given alternatives and write its letter besides the item number in the answer sheet provided

i. Which of the following statements is correct about an atom?
   A. The protons orbit outside the nucleus in shells
   B. The neutrons orbit outside the nucleus in shells
   C. Electrons and neutrons are in the nucleus
   D. Protons and neutrons are in the nucleus
   E. Protons and electrons are in the nucleus

ii. A Chemistry student at Seypalm Secondary School made the following statements:
   (i) Organic chemistry is the Chemistry of carbon compounds.
   (ii) All living things contain carbon compounds
   (iii) Monomers are large molecules made up of polymers
   (iv) Polymers are large molecules obtained from smaller organic molecules
   (v) Carbon is the most unusual atom
   Which of the following is true about the student's statements?
   A. Statement (i) is not correct
   B. Statement (ii) is not correct
   C. Statement (iii) is not correct
   D. Statement (iv) is not correct
   E. All of the student's statements are correct

iii. The repetition of similar chemical properties of elements after a certain specific interval in a given arrangement is termed:
   A. Isotopy
   B. Periodic table
   C. Periodicity

D. General trend
E. Isotopes

iv. Which of the following elements are all located in period 3 of the periodic table?
A. Lithium, Sodium and Potassium
B. Sodium, Magnesium and Chorine
C. Argon, Neon and Xenon
D. Potassium, Calcium and Chlorine
E. Nitrogen, Oxygen and Bromine

v. When determining the effect of concentration of hydrochloric acid in the rate of reaction between the hydrochloric acid and sodium thiosulphate, the rate of the reaction is
A. Independent variable
B. A constant
C. Control experiment
D. Dependent variable
E. Determining factor

vi. This is naturally distilled water; it is almost pure and it contains only gases and dissolved dust. This refers to which of the following?
A. Lake water
B. River water
C. Rain water
D. Spring water
E. Sea water

vii. Bahati weighed exactly 100 grams of anhydrous sodium carbonate using electronic balance. How any moles of oxygen atoms are in the sample that Bahati weighed.
A. 0.03
B. 42.28
C. 1
D. 0.94
E. 2.83

viii. Direct exposure to UV radiations can result in to the following, except:
A. Cholera outbreak
B. Skin cancer
C. Eye damage
D. Skin rash outbreak
E. Impairment of the immune system

ix. The mass of a substance deposited or liberated at an electrode when a current of 1 ampere is passed for 1 second is called:
A. Equivalent weight
B. Electrochemical equivalent
C. Faraday's constant
D. Mole
E. Electrochemical series

x. Contains iron mixed with chromium, nickel and manganese.

A. Stainless steel
B. Bronze
C. Brass
D. Silica gel
E. Galvanizing

2. Match the items in LIST A with the corresponding items in LIST B by writing the letter of the correct response besides the item number in the answer sheet provided

List A	List B
(i) Macromolecules made of many smaller repeating organic molecules.	A. Alkanes
(ii) The most unusual element, whose atom has the ability to join up to itself and form very long chains of atoms.	B. Carbon
(iii) Crude oils and diesel fuels	C. Oxygen
(iv) The fractions consist of mainly methane, ethane, propane, and butane.	D. alkenes
(v) Branches of the main carbon chain, responsible for the unique chemical reactions of a molecule	E. Monomers
(vi) Organic compounds with the same molecular formula but different structural formulae	F. Polymers
	G. Isomers
	H. Natural gas
	I. Functional groups
	J. Petrol

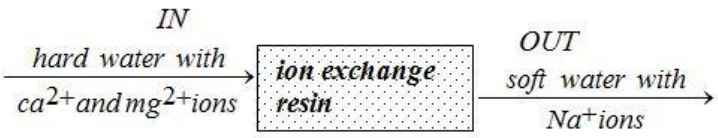
SECTION B (54 MARKS)

3. Chlorine is a dangerous but very useful gas. Verify this observation by:

(a) Explaining how it is dangerous.
(b) Giving four uses of chlorine.

4. (a) Explain the permanent hardness of water with respect to how the condition develops.

(b) The following is an illustration of a method for removing hardness of water, which is known as an ion exchange method:



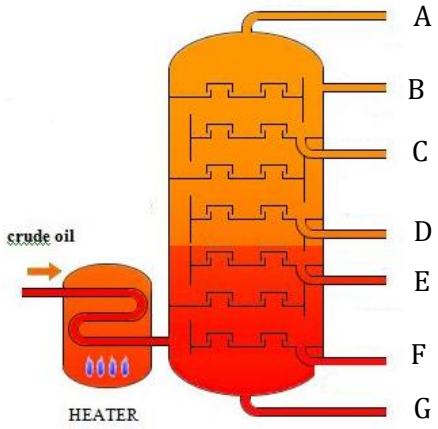
Give more information that corroborates the above illustration in describing the illustrated process of removing hardness of water.

5.(a) State the first and second Faraday's laws of electrolysis.

(b) A steady current of 2 A was passed through a solution containing ions of a divalent metal M (M2+) for nine minutes. The mass of metal M liberated was 0.3552 g.

(i) Calculate the relative atomic mass of metal M.
(ii) Identify the divalent metal M

6.(a) Petroleum (crude oil) is a mixture of several compounds which are separated in the Seypalm refinery by means of apparatus as shown below:



i. What is the name of the apparatus above?
ii. What is the name of the process which is used in separation of crude oil?
iii. What physical property of compounds in the mixture does the separation depend?
iv. Use the letters A to G to describe where the following could be formed:
   a) The fraction that represents gases
   b) The fraction that represents the largest molecules
   c) The fraction that represents liquids with the lowest boiling points
v. State the use of product produced at G

(b) Give the IUPAC systematic names of compounds Q and R

P: $CH_2C(Cl)CH_2CH_2CH_3$

Q: $CH_3CHClCH_2CH_3$

7. Laboratory rules are established in order to avoid accidents, injuries and damages to property; state any nine of them.

8. (a) Define volumetric analysis

(b)What volume of concentrated sulphuric acid is required to make 5 litre of 0.5 M H_2SO_4 solution? The concentrated sulphuric acid is 98% H_2SO_4 by mass and its density is 1.84 g cm^{-3}.

SECTION C (30 marks)

9. Extraction of metals is one of the most important economic activities. The choice of the best method for extraction of a metal from its ore depends on the chemical reactivity of the metal to be extracted. Outline the stages of extraction of moderately reactive metals.

10. Matter is one of the most important topics in the Chemistry subject. The importance of the topic is vividly exhibited when matter changes from one state to another, where life would have been almost impossible or very difficult if matter were unable to change states. In six points, show how far the above argument is valid.

11. The international community is making a lot of effort to curb global warming. However, still there are so many people who seemed not to understand much about global warming, as exhibited by their indifference towards the problem. Explain to those people about the phenomenon, to make them understand how it is caused and its negative effects.